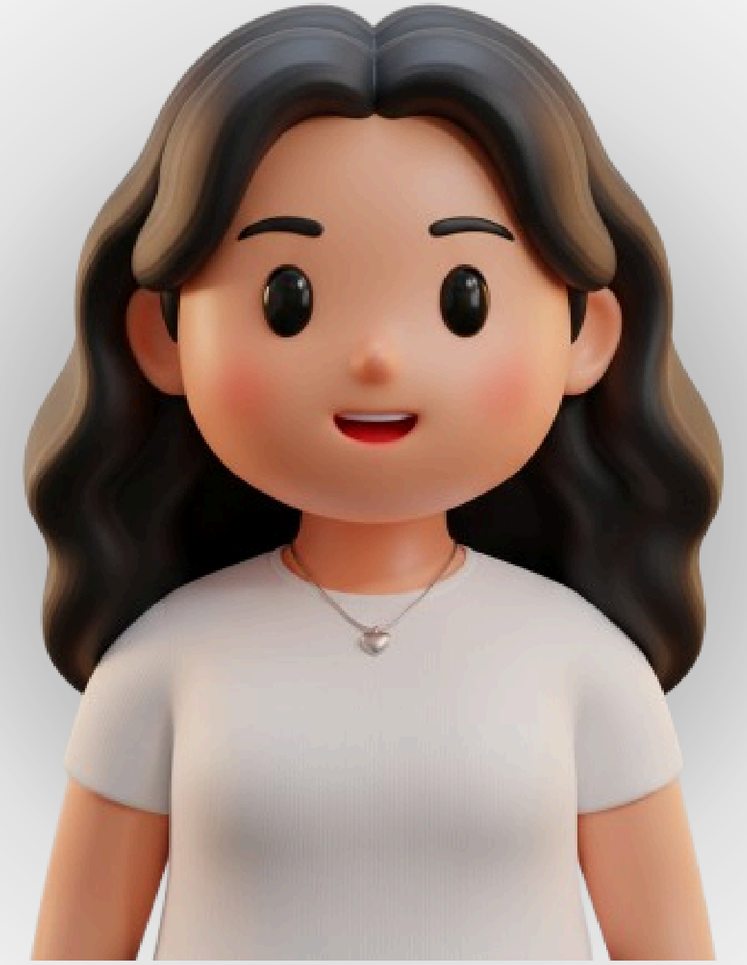
Design Your Desire Interface

Krysthel Portfolio



I Create
User-friendly Designs
That Solve YOUR Problems!

Design Your Desire Interface

ABOUT ME

A Cognitive Science graduate specialising in Human-Computer Interaction and user-centric design. Proven ability to apply cognitive principles and technical skills (Figma, Python, Miro, Android Studio) in user research, data analysis, and prototype development. Eager to create intuitive and efficient solutions that bridge the gap between human capabilities and technology.

Skills

CSS HTML

Access My Full Resume

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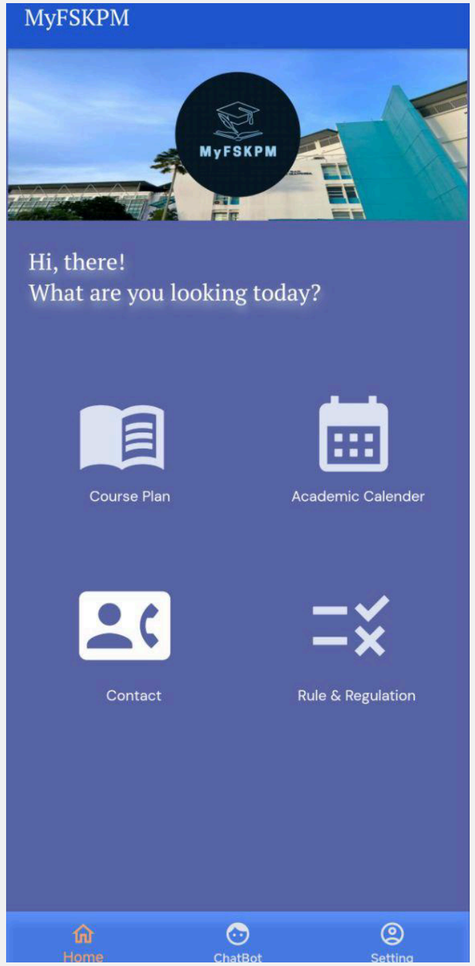
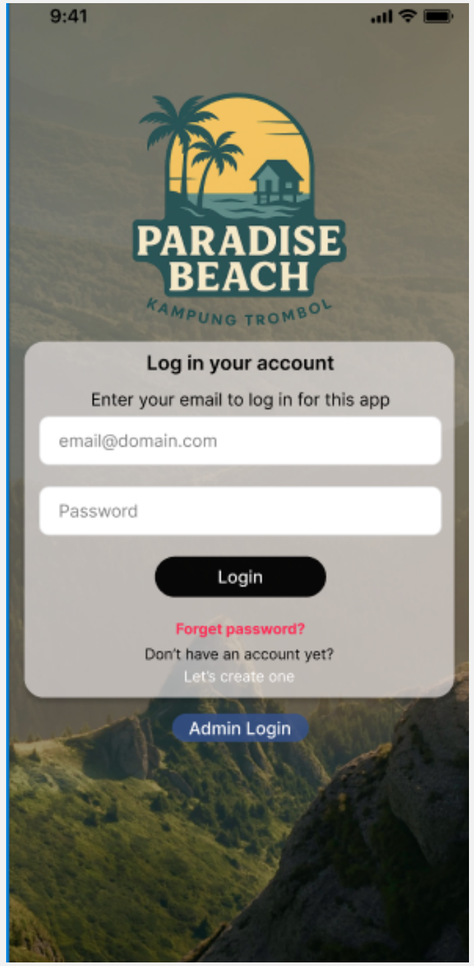
P R O J E C T S





Design Your Desire Interface

MyFSKPM App

Problem

Students at the Faculty of Cognitive Sciences and Human Development (FCSHD) face significant challenges accessing essential academic information. Key information is often buried in lengthy, hard-to-navigate PDF guidebooks or on outdated physical copies. This makes critical tasks, such as course registration and understanding faculty regulations, inefficient and frustrating for students.

Main Objective

The project's primary objective was to design and develop a user-centered mobile application that streamlines student access to faculty guidebooks and essential academic information. The goal was to replace the outdated methods with a solution that provides quick, easy, and reliable access, thereby improving the user experience, enhancing accessibility , and aligning with key usability goals.

Project Type

This was a group project for the KMK3323 Human Computer Interaction course. The project followed an iterative design process, moving from requirements gathering to a paper-based prototype and culminating in a fully functional high-fidelity Android application (APK)

Design Your Desire Interface

MyFSKPM App

Solution

The solution is MyFSKPM, a fully functional, high-fidelity smartphone application. The app consolidates all essential faculty information into one accessible platform, featuring five key modules:

- Course Plan: Provides full course descriptions, prerequisites, credit hours, and elective options.
- Academic Calendar: Offers real-time access to important dates, registration deadlines, and exam schedules.
- Contact Directory: A searchable directory for both lecturer and administrative staff contact details.
- Faculty Regulations: A clear, well-structured section summarizing academic policies, grading criteria, and disciplinary rules.
- Virtual Assistant: An AI chatbot named "Mr FSKPM" integrated to answer student questions about the faculty guidebook

Research

The target user group was identified as students from the Faculty of Cognitive Sciences and Human Development (FCSHD) at UNIMAS, across different programs and years of study.

Course Information

Users need to easily find detailed course descriptions, including prerequisites and credit hours.

Instant Support

Users need a way to get quick answers to common questions regarding faculty information, which was met with the "Mr FSKPM" chatbot.

Contacts

Users need to quickly find contact details for lecturers and faculty staff

Rules

Users need a clear summary of faculty regulations, such as attendance requirements and grading criteria.

Schedules

Users need a reliable, updated academic calendar for deadlines and exam schedules.

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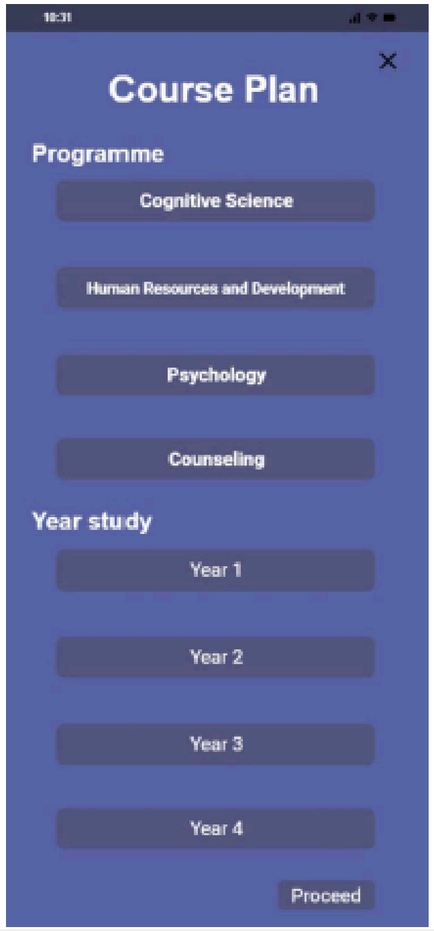
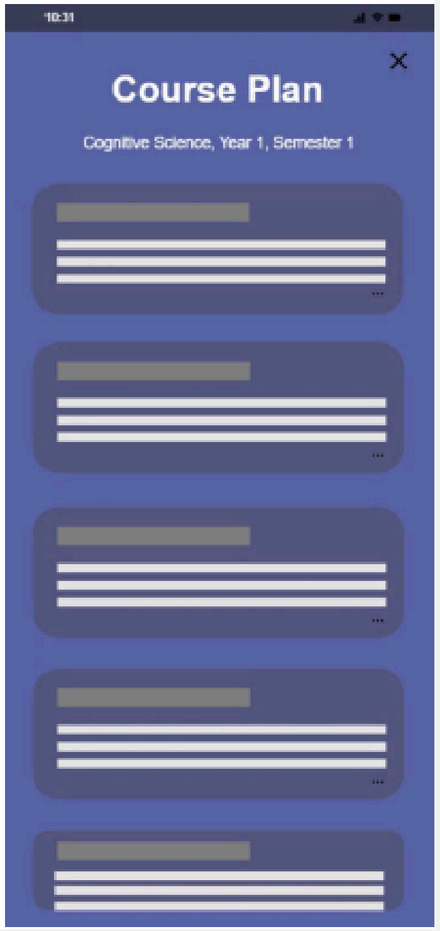
DESIGN

Low-Fidelity

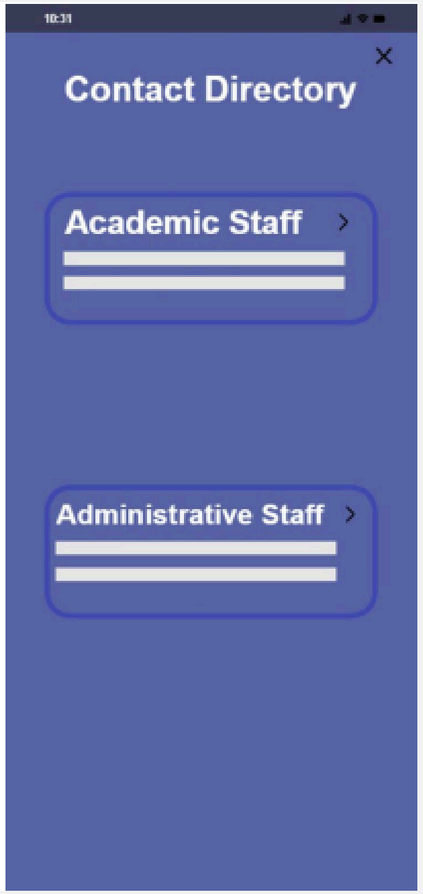




Design Your Desire Interface

DESIGN

Low-Fidelity



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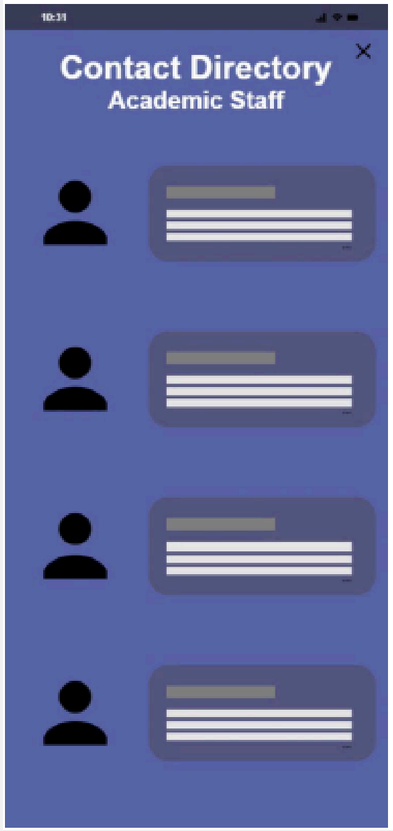
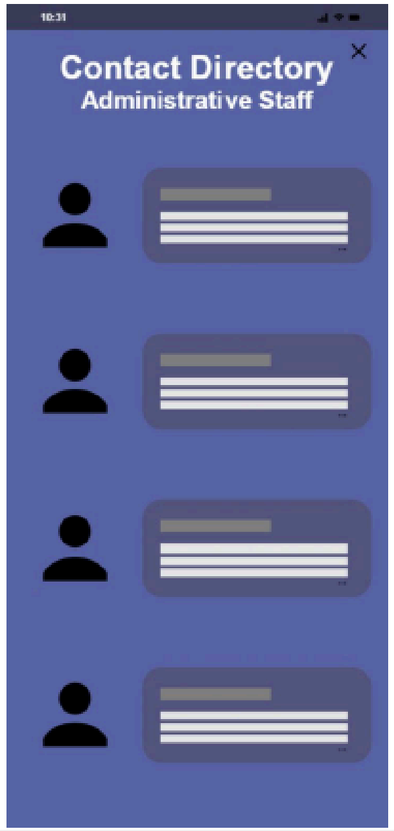
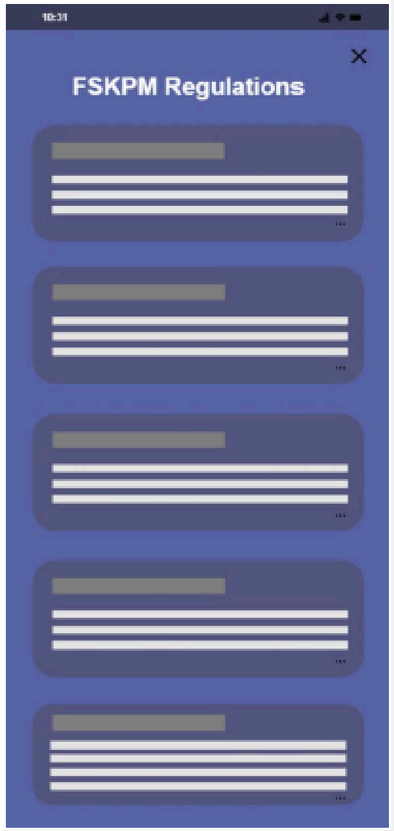
DESIGN

Low-Fidelity



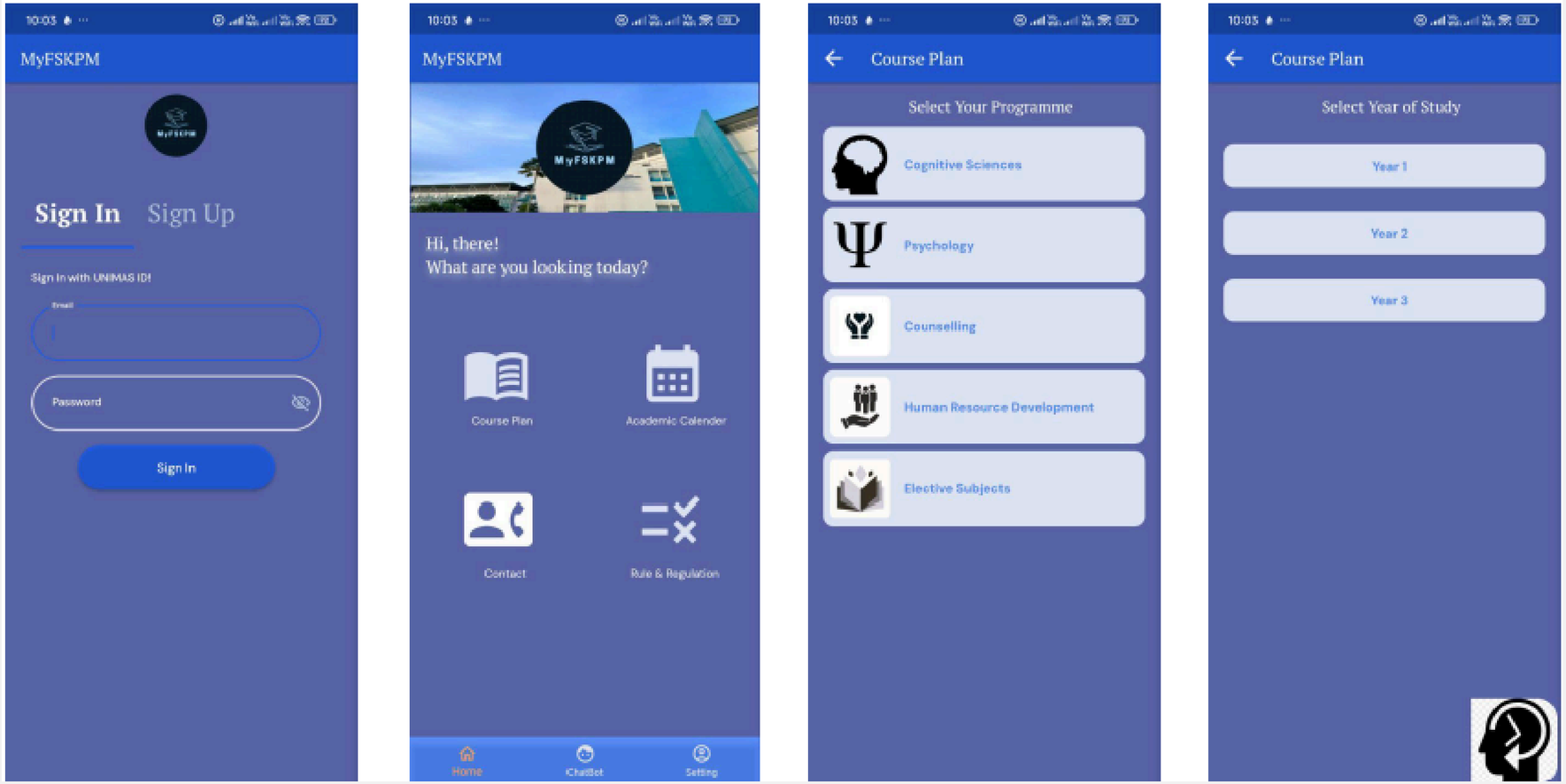




Design Your Desire Interface

DESIGN

High-Fidelity (Fully Functional App)



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High-Fidelity (Fully Functional App)

Design Your Desire Interface

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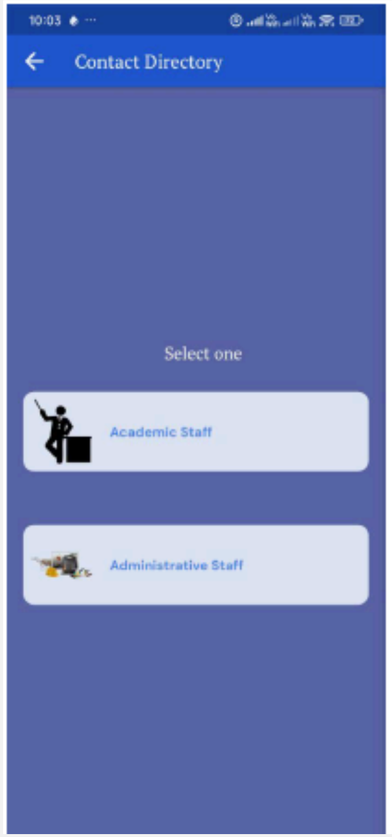
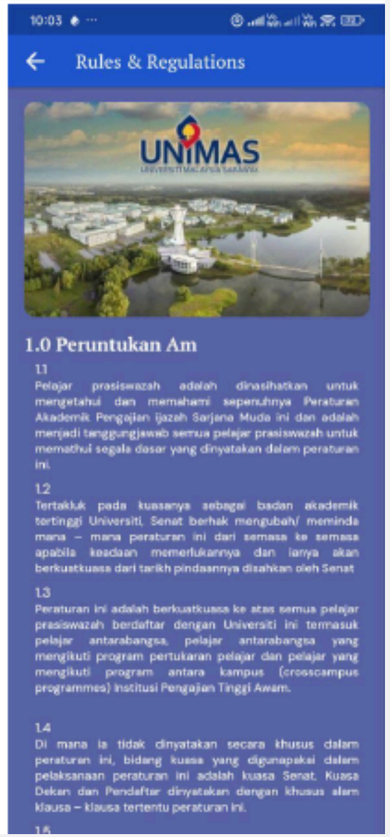
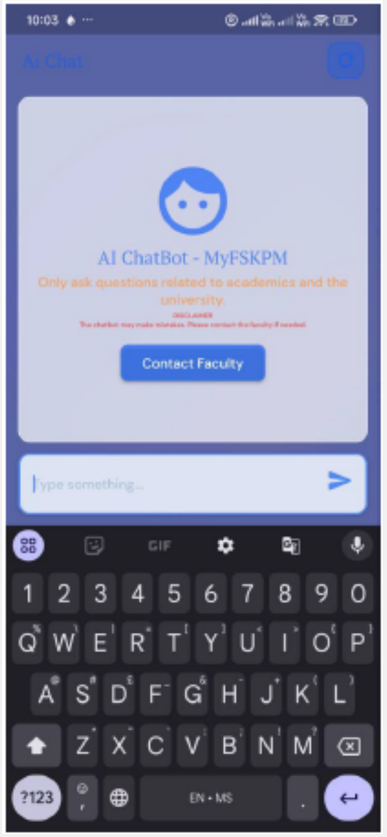
High-Fidelity (Fully Functional App)







User Testing & Evaluation

Low-Fidelity Evaluation

- Method: A cooperative evaluation was conducted , as per the project requirements.
- Participants: Two students from the target faculty (FCSHD).
- Tasks: Participants were asked to locate key features (like the Course Plan and Chatbot) and provide open-ended feedback on the design.
- Findings: While users found the icons intuitive and the layout "orderly" , the evaluation successfully identified several key areas for improvement: the search bar was deemed "unnecessary," a navigation bar was missing, an authentication system was recommended, and the course plan needed better organization

User Testing & Evaluation

High-Fidelity Usability Testing

- Method: Formal usability testing was conducted on the fully functional app.
- Participants: Six FCSHD students from various programs and years of study.
- Benchmark Tasks: Participants were given five specific tasks, such as "Find the course details for KMF1014," "Find the details for March 15 in the Academic Calendar," and "Use the AI ChatBot".
- Goals & Metrics: Success was measured against five usability criteria: Learnability, Efficiency, Memorability, Error rate, and Satisfaction .
- Findings: The high-fidelity prototype was completely successful.
- Learnability: All participants completed tasks independently on their first try.
- Efficiency: All tasks were completed in less than 1 minute, well under the 3-minute benchmark, demonstrating high efficiency.
- Error Rate: No significant errors were observed for any participant.
- Satisfaction: All participants reported a positive experience, describing the app as "smooth, easy to navigate, and free of issues". No user frustration was reported. The only minor suggestion was from one user who proposed adjusting the theme colors

Design Your Desire Interface

Kampung Trombol App

Problem

Kampung Trombol is a unique beach destination, but it lacks a modern, user-friendly digital platform to promote its attractions, manage services, and engage with visitors. An effective mobile presence is needed to streamline the visitor experience and provide an administrative tool for managing the location's offerings.

Main Objective

This project was a two-phase endeavor. The first objective was to design a complete, high-fidelity UI skeleton for the "Trombol Paradise Beach" mobile application using Figma. The second objective was to convert this UI prototype into a fully functional, database-driven mobile application using the Flutter SDK and Firebase backend.

Solution

The solution is a comprehensive, cross-platform mobile application built with Flutter. The app serves two distinct user types: Customers and Admins.

- For Customers: The app provides a seamless interface to browse and book offerings. Users can explore Activities , Services (like beach stays) , local Products , and Dining options. It includes full user authentication (registration, login, password recovery) , a multi-item booking cart , a dummy payment/checkout process , and an AI-powered chatbot ("Drake") for 24/7 customer support.

- For Admins: After a separate admin login, administrators gain access to a full content management system. They have complete CRUD (Create, Read, Update, Delete) capabilities over all app content, including adding new activities, editing product prices, uploading images, and deleting services. They can also manage user accounts.

- Technology: The app uses Firebase for all backend services, including Firebase Authentication for login , Firebase Firestore as the NoSQL database , and Firebase Storage for image uploads. The AI chatbot is powered by Google's Gemini 2.0 Flash model.

User Research & Needs

While the project did not involve direct user interviews, the requirements were established through project briefs and competitive analysis. The team analyzed other tourism-focused applications, such as the Swahili Beach Resort app, to identify essential features and user-friendly navigation patterns.

Admin Needs

- A secure, separate login.
- The ability to add, edit, and delete all content (activities, services, products, dining).
- An interface for uploading corresponding images for all content.
- The ability to manage user details, including updating or deleting accounts

Customer Needs

- Standard user authentication (login, registration, forgot password).
- An intuitive homepage to browse all available offerings.
- The ability to select packages, add them to a cart, and select a date.
- A simple checkout and payment (dummy) process.
- A way to manage their own profile and change passwords.
- Access to customer support.

Design Process & High-Fidelity Prototype

A distinct brand identity was created for "Paradise Beach Kampung Trombol".

- Logo: The logo features coconut trees, a house on the water, the sea, and a sunset, capturing the village and beach atmosphere.
- Color Palette: The primary colors are golden yellow, Hex: #F7C548 (representing sunshine and tourism) and teal green, Hex: #306E72 and dark olive green, Hex: #22484C (reflecting nature and tranquility).

Link and QR code to Low-Fidelity Prototype

https://www.figma.com/proto/3I74gCaayHDbrHQD62kB2W/Paradise-Beach?node-id=418-553&t=DXvwNDHdPPIsZqrS-1

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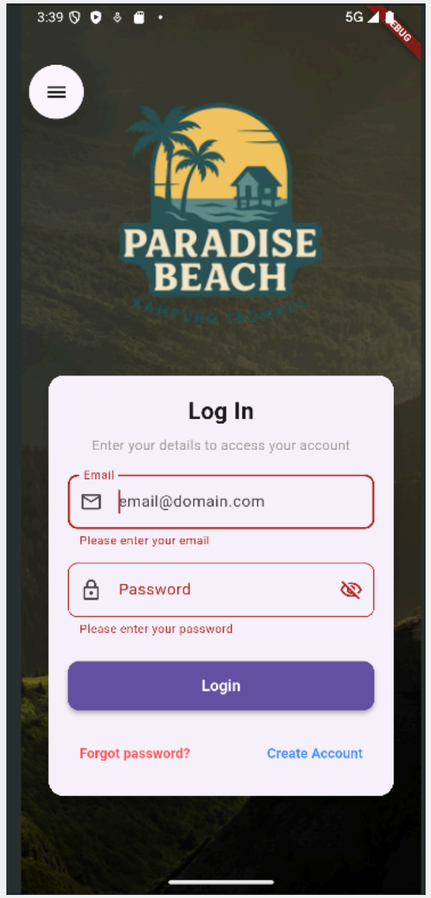
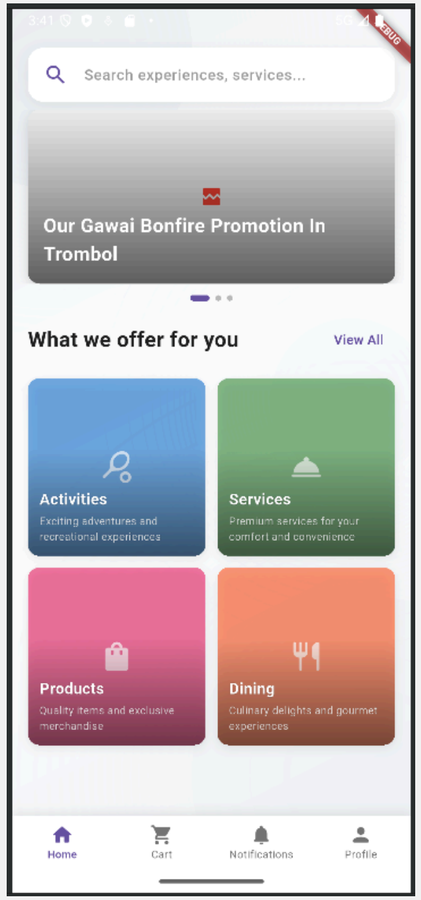
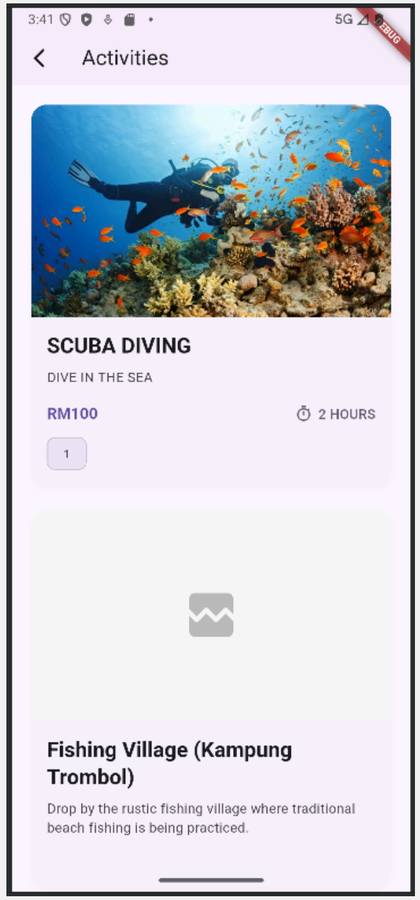
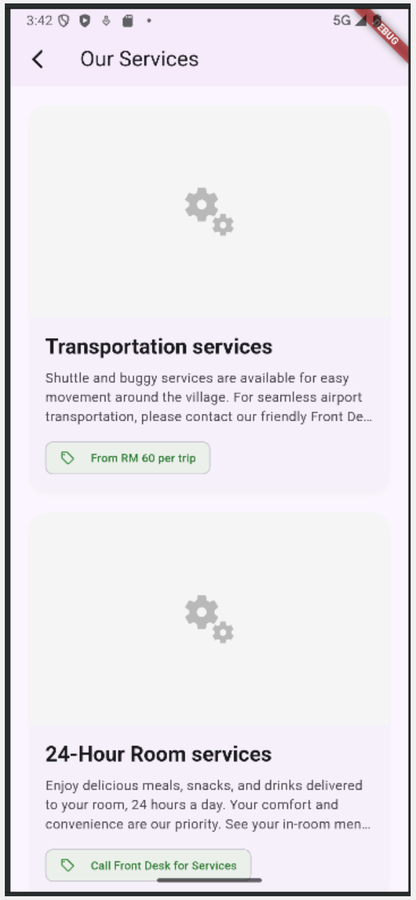
High-Fidelity (Fully Functional App)









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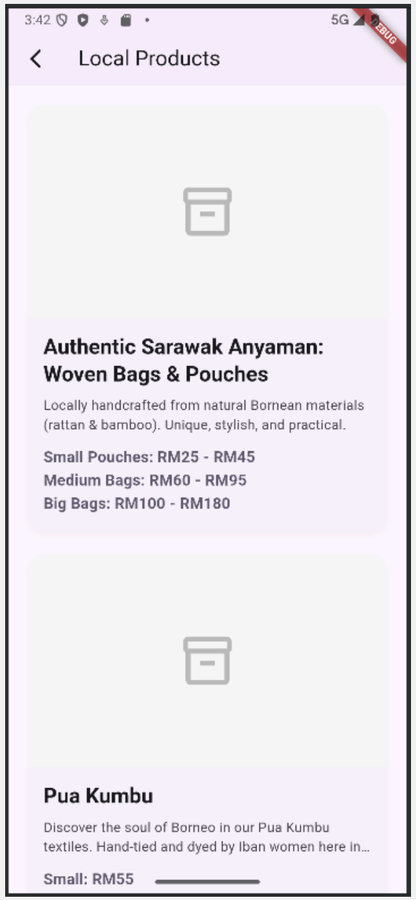

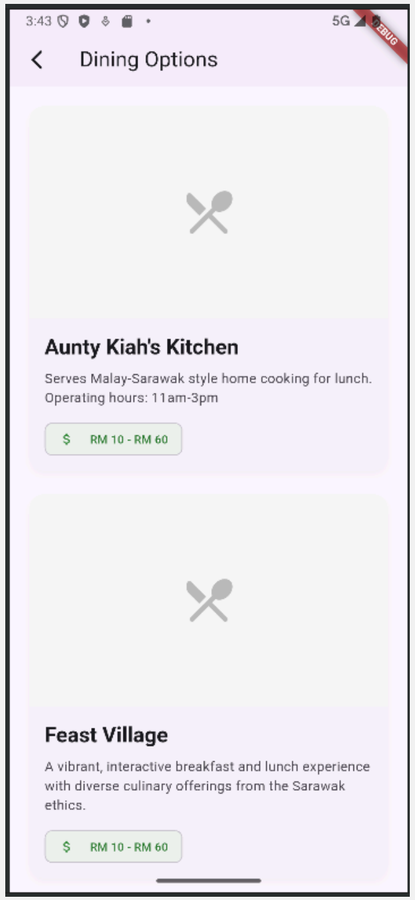

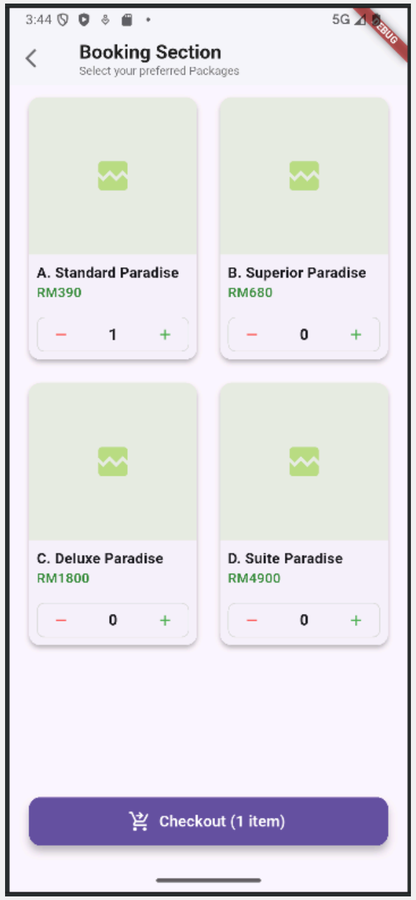

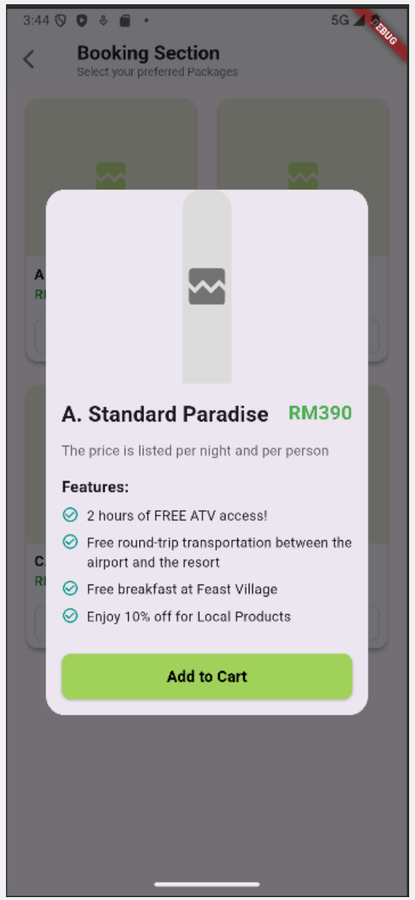

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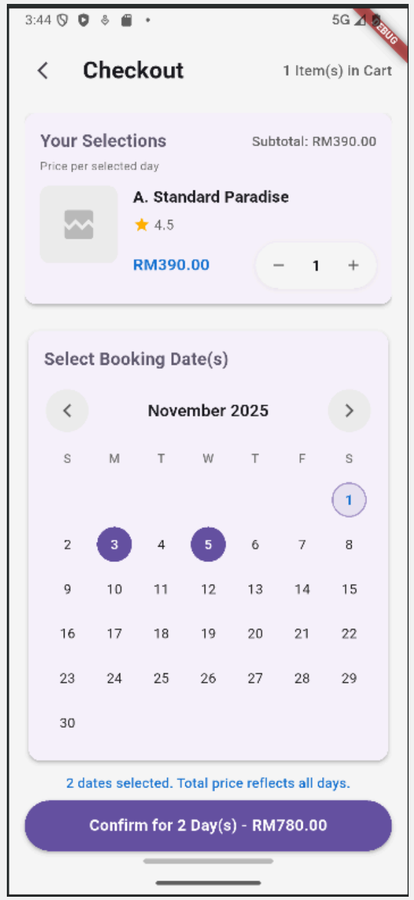

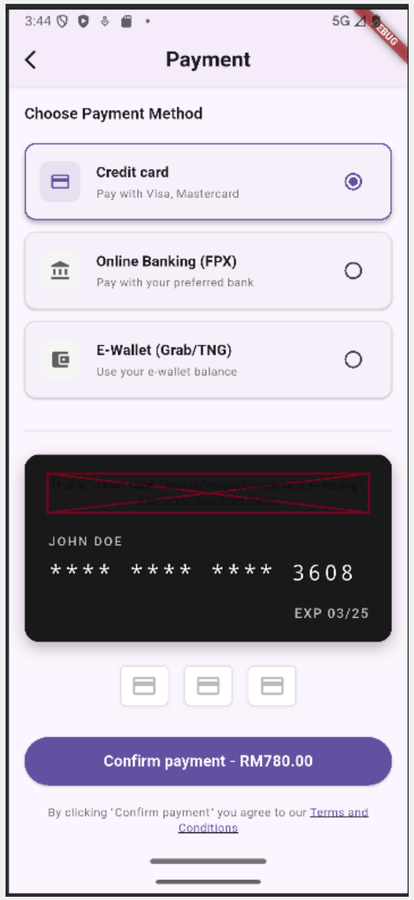

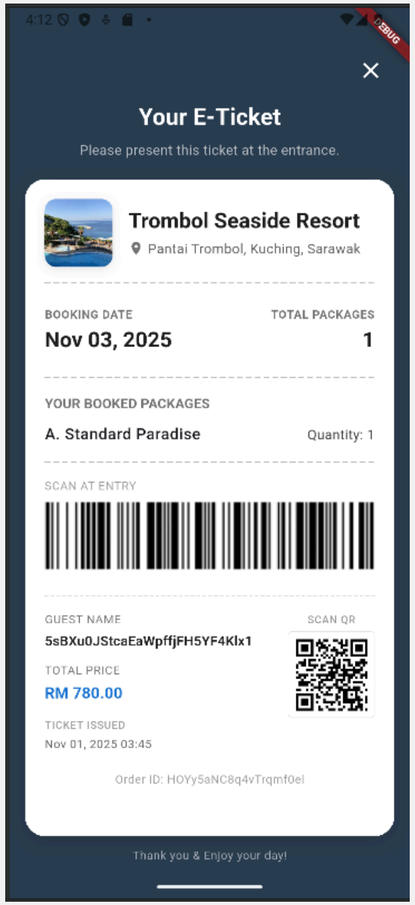

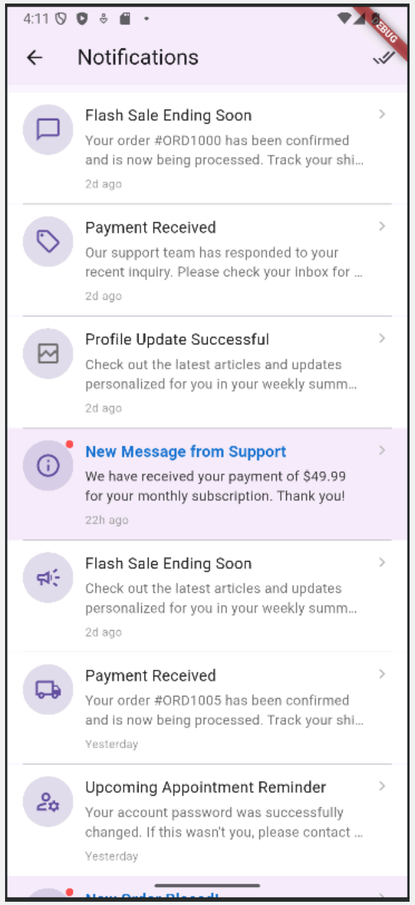

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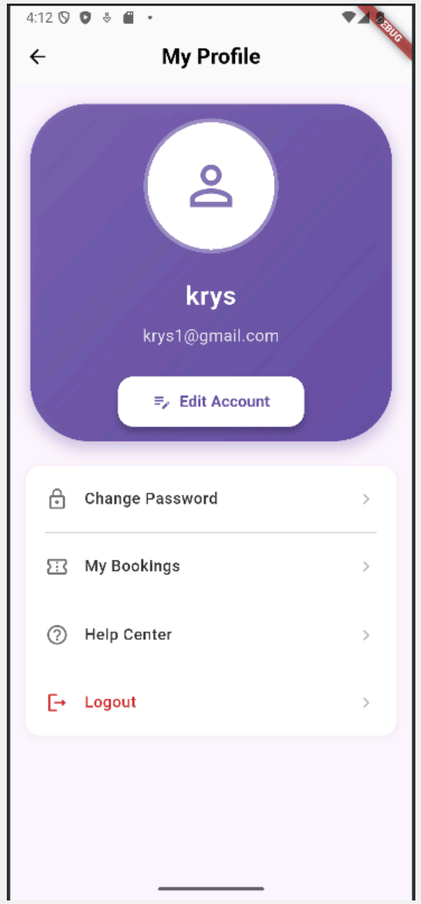

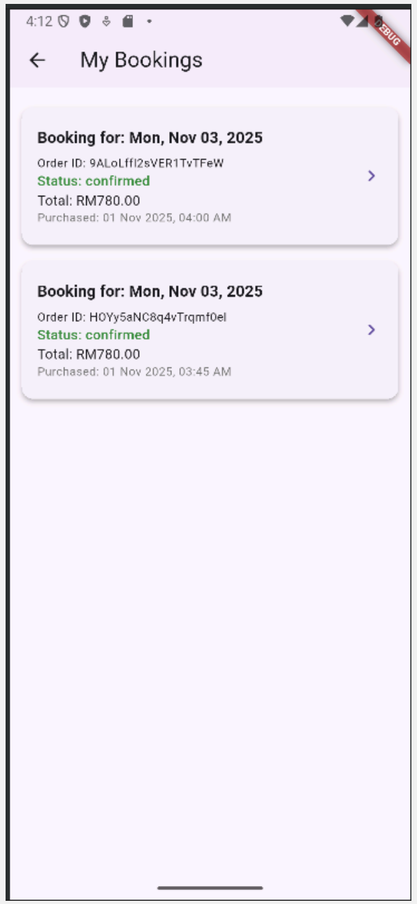

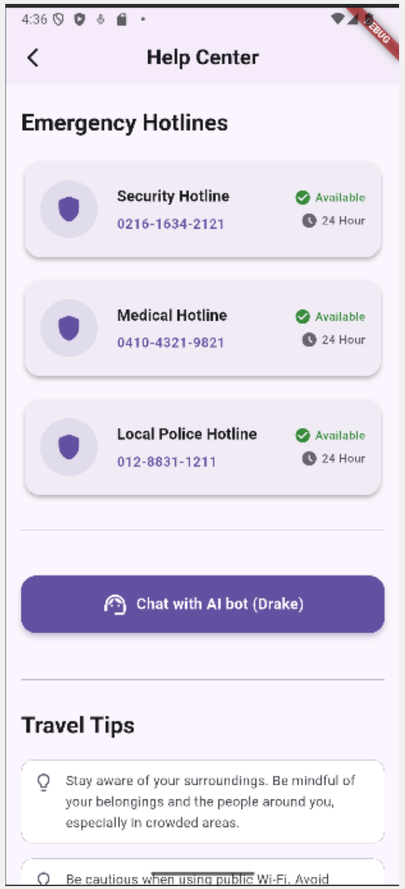

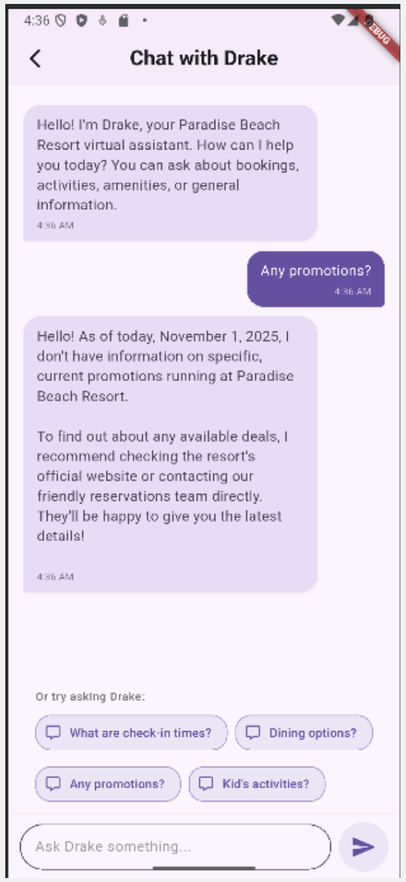

Let's Work Together!